



॥ त्वं ज्ञानमयो विज्ञानमयोऽसि ॥

भारतीय प्रौद्योगिकी संस्थान जोधपुर

एनएच 65, सुरपुरा बाईपास रोड, करवड़, राजस्थान 342037

Indian Institute of Technology Jodhpur

NH 65, Surpura Bypass Rd, Karwar, Rajasthan 342037

Course: Engineering Mechanics

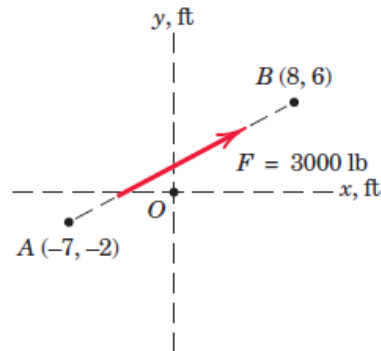
Code: MEL1010

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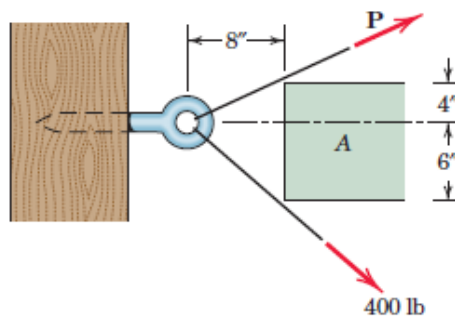
Tutorial: 01

**Question 1:** If a man weighs 155 lb on earth, specify (a) his mass in slugs, (b) his mass in kilograms, and (c) his weight in newton. If the man is on the moon, where the acceleration due to gravity is  $K_m = 5.30 \text{ ft/s}^2$ , determine (d) his weight in pounds, and (e) his mass in kilograms.

**Question 2:** The line of action of the 3000-lb force runs through the points A and B as shown in the figure. Determine the  $x$  and  $y$  scalar components of  $F$ .



**Question 3:** It is desired to remove the spike from the timber by applying force along its horizontal axis. An obstruction A prevents direct access, so that two forces, one 400 lb and the other  $P$ , are applied by cables as shown. Compute the magnitude of  $P$  necessary to ensure a resultant  $T$  directed along the spike. Also find  $T$ .



**Question 4:** Determine the moment about point A of each of the three forces acting on the beam.



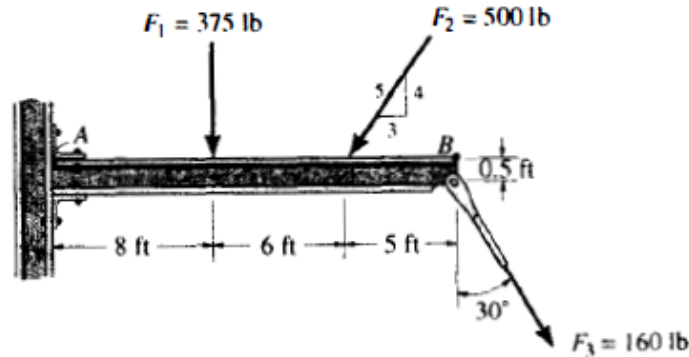
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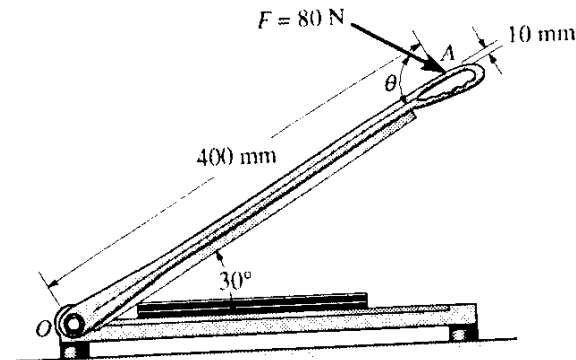
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**Question 5:** A force of 80 N acts on the handle of the paper cutter at A. Determine the moment created by this force about the hinge at O, if  $\theta = 60^\circ$ . At what angle  $\theta$  should the force be applied so that the moment it creates about point O is maximum (clockwise)? What is this maximum moment?



**Question 6:** In the design of the robot to insert the small cylindrical part into a close-fitting circular hole, the robot arm must exert a 90-N force P on the part parallel to the axis of the hole as shown. Determine the components of the force which the part exerts on the robot along axes (a) parallel and perpendicular to the arm AB, and (b) parallel and perpendicular to the arm BC.



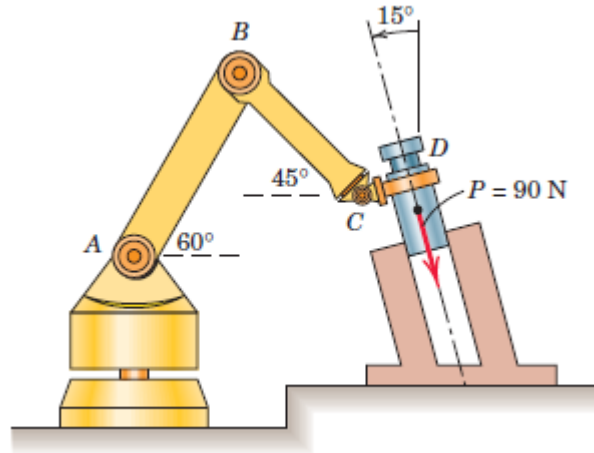
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**Question 7:** Watch the following video to understand the inception of Metric measuring system: <https://www.youtube.com/watch?v=nRnuY1Vao0o&t=1026s>



Measurement System

**Question 8:** Read and explore the history of measurement systems in India.